



Submission Deadline: 28/09/26	Numerical Analysis ES341 Fall 2026	Practice Assignment 1
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2. Find intervals containing solutions to the following equations.
 - a. $x - 3^{-x} = 0$
 - b. $4x^2 - e^x = 0$
 - c. $x^3 - 2x^2 - 4x + 2 = 0$
 - d. $x^3 + 4.001x^2 + 4.002x + 1.101 = 0$
3. Show that $f'(x)$ is 0 at least once in the given intervals.
 - a. $f(x) = 1 - e^x + (e - 1) \sin((\pi/2)x)$, $[0, 1]$
 - b. $f(x) = (x - 1) \tan x + x \sin \pi x$, $[0, 1]$
 - c. $f(x) = x \sin \pi x - (x - 2) \ln x$, $[1, 2]$
 - d. $f(x) = (x - 2) \sin x \ln(x + 2)$, $[-1, 3]$
4. Find $\max_{a \leq x \leq b} |f(x)|$ for the following functions and intervals.
 - a. $f(x) = (2 - e^x + 2x)/3$, $[0, 1]$
 - b. $f(x) = (4x - 3)/(x^2 - 2x)$, $[0.5, 1]$
 - c. $f(x) = 2x \cos(2x) - (x - 2)^2$, $[2, 4]$
 - d. $f(x) = 1 + e^{-\cos(x-1)}$, $[1, 2]$
7. Let $f(x) = x^3$.
 - a. Find the second Taylor polynomial $P_2(x)$ about $x_0 = 0$.
 - b. Find $R_2(0.5)$ and the actual error in using $P_2(0.5)$ to approximate $f(0.5)$.
 - c. Repeat part (a) using $x_0 = 1$.
 - d. Repeat part (b) using the polynomial from part (c).
8. Find the third Taylor polynomial $P_3(x)$ for the function $f(x) = \sqrt{x+1}$ about $x_0 = 0$. Approximate $\sqrt{0.5}$, $\sqrt{0.75}$, $\sqrt{1.25}$, and $\sqrt{1.5}$ using $P_3(x)$, and find the actual errors.
9. Find the second Taylor polynomial $P_2(x)$ for the function $f(x) = e^x \cos x$ about $x_0 = 0$.
 - a. Use $P_2(0.5)$ to approximate $f(0.5)$. Find an upper bound for error $|f(0.5) - P_2(0.5)|$ using the error formula, and compare it to the actual error.
 - b. Find a bound for the error $|f(x) - P_2(x)|$ in using $P_2(x)$ to approximate $f(x)$ on the interval $[0, 1]$.

13. Find the fourth Taylor polynomial $P_4(x)$ for the function $f(x) = xe^{x^2}$ about $x_0 = 0$.
- Find an upper bound for $|f(x) - P_4(x)|$, for $0 \leq x \leq 0.4$.
 - Approximate $\int_0^{0.4} f(x) dx$ using $\int_0^{0.4} P_4(x) dx$.
 - Find an upper bound for the error in (b) using $\int_0^{0.4} P_4(x) dx$.
 - Approximate $f'(0.2)$ using $P'_4(0.2)$, and find the error.
5. Use the Bisection method to find solutions accurate to within 10^{-5} for the following problems.
- $x - 2^{-x} = 0$ for $0 \leq x \leq 1$
 - $e^x - x^2 + 3x - 2 = 0$ for $0 \leq x \leq 1$
 - $2x \cos(2x) - (x + 1)^2 = 0$ for $-3 \leq x \leq -2$ and $-1 \leq x \leq 0$
 - $x \cos x - 2x^2 + 3x - 1 = 0$ for $0.2 \leq x \leq 0.3$ and $1.2 \leq x \leq 1.3$
6. Use the Bisection method to find solutions, accurate to within 10^{-5} for the following problems.
- $3x - e^x = 0$ for $1 \leq x \leq 2$
 - $2x + 3 \cos x - e^x = 0$ for $0 \leq x \leq 1$
 - $x^2 - 4x + 4 - \ln x = 0$ for $1 \leq x \leq 2$ and $2 \leq x \leq 4$
 - $x + 1 - 2 \sin \pi x = 0$ for $0 \leq x \leq 0.5$ and $0.5 \leq x \leq 1$
3. The following four methods are proposed to compute $21^{1/3}$. Rank them in order, based on their apparent speed of convergence, assuming $p_0 = 1$.
- $p_n = \frac{20p_{n-1} + 21/p_{n-1}^2}{21}$
 - $p_n = p_{n-1} - \frac{p_{n-1}^3 - 21}{3p_{n-1}^2}$
 - $p_n = p_{n-1} - \frac{p_{n-1}^4 - 21p_{n-1}}{p_{n-1}^2 - 21}$
 - $p_n = \left(\frac{21}{p_{n-1}}\right)^{1/2}$
4. The following four methods are proposed to compute $7^{1/5}$. Rank them in order, based on their apparent speed of convergence, assuming $p_0 = 1$.
- $p_n = p_{n-1} \left(1 + \frac{7 - p_{n-1}^5}{p_{n-1}^2}\right)^3$
 - $p_n = p_{n-1} - \frac{p_{n-1}^5 - 7}{p_{n-1}^2}$
 - $p_n = p_{n-1} - \frac{p_{n-1}^5 - 7}{5p_{n-1}^4}$
 - $p_n = p_{n-1} - \frac{p_{n-1}^5 - 7}{12}$
14. Use a fixed-point iteration method to determine a solution accurate to within 10^{-4} for $x = \tan x$, for x in $[4, 5]$.
15. Use a fixed-point iteration method to determine a solution accurate to within 10^{-2} for $2 \sin \pi x + x = 0$ on $[1, 2]$. Use $p_0 = 1$.